

Astra-Veritas Module Specification

AI-Assisted Verification & Truth-Attestation Engine

National Sovereignty & Technology Trust (NSTT) Governance Suite

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Revision History

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1. Module Overview & Purpose

1.1 Astra-Veritas Identity and Mission Statement

1.1.1The Astra-Veritas module is the designated AI-assisted verification and truth-attestation engine of the National Sovereignty & Technology Trust (NSTT) governance suite. It operates as a deterministic, tamper-evident subsystem responsible for the continuous verification of governance compliance, information integrity, operational fidelity, and ZHC-001 alignment across the entirety of NSTT operations.

1.1.2 Mission Statement: Astra-Veritas shall provide an unimpeachable, deterministic record of truth attestation across all NSTT-governed processes, documents, systems, and personnel actions. It shall operate without ambiguity, bias, or discretionary exception, ensuring that every attestation issued under its authority constitutes an authoritative verification event admissible as audit evidence under NSTT-GOV-AUD-001.

1.1.3The module derives its name from the Latin *astra* (stars — immutable points of reference) and *veritas* (truth — the foundational value of governance integrity), reflecting its mandate to serve as a fixed, reliable point of reference within the NSTT governance constellation.

1.2 Role Within the NSTT Governance Ecosystem

1.2.1Astra-Veritas occupies the verification layer of the NSTT governance architecture. It receives inputs from, and issues attestation outputs to, all primary NSTT governance modules, including Blue Bird (strategic operations), Moonlight (data sovereignty and classification), and the sovereign compliance framework governed by NSTT-GOV-SOV-001.

1.2.2The module does not direct or authorize governance decisions. Its function is strictly verificatory and attestatory: it confirms the factual and procedural correctness of governance events, generates cryptographically anchored attestation certificates, and flags anomalies for escalation to designated governance authorities.

1.2.3Astra-Veritas serves as the single authoritative source of truth attestation within the NSTT ecosystem. No governance record, operational decision record, or compliance declaration shall be considered verified in the absence of a valid Astra-Veritas attestation certificate.

1.3 ZHC-001 Verification Mandate

1.3.1Astra-Veritas is the primary verification engine for ZHC-001 compliance events within the NSTT governance suite. All ZHC-001 verification events shall be processed, attested, and logged exclusively through the Astra-Veritas module.

1.3.2The ZHC-001 verification mandate requires Astra-Veritas to perform real-time compliance signal assessment, integrity hash validation, and deterministic output generation for every ZHC-001 event registered within the NSTT operational environment.

1.3.3ZHC-001 compliance verification results produced by Astra-Veritas shall be treated as primary evidence in all NSTT audit proceedings and shall supersede all other compliance declarations unless formally challenged through the governance dispute resolution procedure specified in NSTT-GOV-LEGAL-001.

1.4 Deterministic Governance Support Function

1.4.1All verification outputs produced by Astra-Veritas are deterministic: given identical inputs under identical operational conditions, the module shall produce identical verification results. This property is essential to the evidentiary reliability of attestation certificates and the reproducibility of audit findings.

1.4.2Deterministic operation is enforced through the use of immutable verification logic, version-locked rule sets, and integrity-hashed governance policy snapshots. Any modification to the verification logic requires a full module version increment and re-authorization per Section 13 of this specification.

2. Module Definitions

The following definitions apply exclusively within the context of this specification and all documents cross-referenced herein. Where a term is defined in NSTT-GOV-

MASTER-001, that definition shall govern; definitions herein are supplemental and module-specific.

Term	Definition
Astra-Veritas Engine	The primary AI-assisted verification processing core of the Astra-Veritas module, responsible for executing deterministic verification logic against governance inputs.
Verification Node	A designated logical endpoint within the NSTT architecture at which verification events are initiated, processed, or recorded by the Astra-Veritas Engine.
Truth Attestation	The formal, module-certified declaration that a specific governance artifact, event, or state conforms to its specified governance parameters as determined by the Astra-Veritas Engine.
Integrity Hash	A cryptographically generated fixed-length digest of a governance artifact or operational record, used to verify that the artifact has not been altered since the hash was computed.
ZHC-001 Verification Event	Any governance event that triggers a mandatory ZHC-001 compliance check, including policy activations, cross-module data exchanges, and sovereign boundary interactions as defined in ZHC-001.
Compliance Signal	A structured output from the Astra-Veritas Engine indicating the compliance status (COMPLIANT, NON-COMPLIANT, PENDING, or INDETERMINATE) of a given governance event relative to its applicable governance standard.
Anomaly Flag	A system-generated alert raised by the Anomaly Detection Subsystem indicating a deviation from expected governance parameters, integrity baseline, or ZHC-001 compliance thresholds.
Deterministic Verification	A verification process in which the output is uniquely and reproducibly determined by the inputs and the version-locked verification logic, with no stochastic or probabilistic components.
Attestation Certificate	A formally structured, module-signed digital document issued by the Attestation Certificate Generator confirming the result of a verification event, including its classification, scope, integrity hash, and

Term	Definition
	chain of custody reference.

3. Architecture Overview

3.0.1The Astra-Veritas module is composed of six integrated subsystems operating within a layered, deterministic verification architecture. The subsystems communicate through authenticated, integrity-hashed internal messaging channels. No subsystem may modify the output of another subsystem's primary processing function. The architecture is described below in logical order from input to output.

Architecture Topology – Logical Flow

External Inputs → [Governance API Interface] → [Core Verification Engine] → [ZHC-001 Integration Layer] → [Anomaly Detection Subsystem] → [Attestation Certificate Generator] → [Audit Trail Logging Module] → Output Consumers (NSTT modules, AUD-001, SOV-001)

3.1 Core Verification Engine

Subsystem: Core Verification Engine (CVE)

Purpose: The CVE is the primary deterministic processing unit of Astra-Veritas. It executes all verification logic against governance inputs and produces primary compliance signals.

Inputs: Governance artifacts (documents, records, policy snapshots), process event streams, personnel identity tokens, system state reports, on-demand verification requests.

Outputs: Compliance Signals (COMPLIANT / NON-COMPLIANT / PENDING /

INDETERMINATE), integrity hash records, verification event logs.

ZHC-001 Interface: The CVE consults the ZHC-001 Integration Layer for every verification event tagged as a ZHC-001 Verification Event. ZHC-001 compliance status is a mandatory component of all CVE output records for such events.

3.1.1The CVE operates on version-locked rule sets. Rule set updates require formal authorization per Section 13.2 and must be accompanied by a new module version increment. The active rule set version is embedded in every CVE output record.

3.2 ZHC-001 Integration Layer

Subsystem: ZHC-001 Integration Layer (ZIL)

Purpose: The ZIL provides a dedicated, authenticated integration interface between the Astra-Veritas Engine and the ZHC-001 governance framework. It translates ZHC-001 policy parameters into machine-actionable verification criteria and returns structured ZHC-001 compliance verdicts to the CVE.

Inputs: ZHC-001 event triggers, current ZHC-001 policy snapshot (integrity-hashed), CVE verification requests.

Outputs: ZHC-001 compliance verdicts, ZHC-001 policy deviation reports, escalation flags for override conditions.

ZHC-001 Interface: The ZIL is the sole authorized conduit for all ZHC-001 communication within Astra-Veritas. It maintains a real-time authenticated session with the ZHC-001 policy registry and reports any session integrity failure as a Severity 1 Anomaly Flag.

3.3 Attestation Certificate Generator

Subsystem: Attestation Certificate Generator (ACG)

Purpose: The ACG produces formally structured Attestation Certificates based on CVE and ZIL outputs. It assigns certificate classifications, embeds integrity hashes, records chain of custody metadata, and applies the module's cryptographic signature.

Inputs: Verified CVE compliance signals, ZIL ZHC-001 verdicts, attestation request parameters, certificate type classification rules.

Outputs: Signed Attestation Certificates (see Section 6), certificate issuance records, revocation registry updates.

ZHC-001 Interface: All Attestation Certificates produced for ZHC-001 Verification Events are classified as ZHC-001 Attestation Certificates and carry the ZHC-001 integration reference in their header.

3.4 Anomaly Detection Subsystem

Subsystem: Anomaly Detection Subsystem (ADS)

Purpose: The ADS monitors all CVE and ZIL processing streams in real time for deviations from expected governance parameters, integrity baselines, and ZHC-001 compliance thresholds. It raises Anomaly Flags and initiates automated response actions per the Anomaly Classification Matrix (Section 7.1).

Inputs: CVE processing stream, ZIL compliance verdicts, Audit Trail Logging Module outputs, external system state indicators.

Outputs: Anomaly Flags (classified by severity), automated response commands, escalation notifications, investigation dossier initiation requests.

ZHC-001 Interface: Any anomaly detected within a ZHC-001 Verification Event processing stream is automatically classified at a minimum Severity 2 and reported to the ZIL for ZHC-001 escalation handling.

3.5 Governance API Interface

Subsystem: Governance API Interface (GAI)

Purpose: The GAI provides the authenticated, access-controlled external interface through which all NSTT modules, authorized personnel, and integrated governance systems submit verification requests and receive attestation outputs from Astra-Veritas.

Inputs: Authenticated API requests (on-demand verification, status queries, certificate retrieval), module-to-module integration calls from Blue Bird, Moonlight, and SOV-001.

Outputs: Verification request acknowledgements, compliance signal summaries, Attestation Certificate delivery, anomaly notification dispatches.

ZHC-001 Interface: The GAI enforces ZHC-001 access classification for all inbound requests. Requests originating from ZHC-001-designated sovereign interaction points are tagged and routed through the ZIL before CVE processing.

3.6 Audit Trail Logging Module

Subsystem: Audit Trail Logging Module (ATLM)

Purpose: The ATLM creates a tamper-evident, append-only log of every verification event, attestation issuance, anomaly flag, and system state transition within Astra-Veritas. Logs are maintained in compliance with NSTT-GOV-AUD-001 evidence retention standards.

Inputs: All CVE, ZIL, ACG, ADS, and GAI event streams; module-level system state events.

Outputs: Structured audit log entries (append-only), integrity-chained log manifests, AUD-001 evidence packages, periodic log integrity verification reports.

ZHC-001 Interface: ZHC-001 Verification Event log entries are maintained in a dedicated, separately integrity-hashed ZHC-001 audit partition, accessible only to authorized personnel under NSTT-GOV-AUD-001.

4. ZHC-001 Integration Specification

4.1 ZHC-001 Event Triggers Within Astra-Veritas

4.1.1A ZHC-001 Verification Event is triggered within Astra-Veritas upon the occurrence of any of the following conditions:

- Activation or modification of a ZHC-001-governed policy parameter within the NSTT operational environment;
- Cross-sovereign-boundary data exchange involving NSTT-classified assets;
- Issuance or revocation of a governance authorization affecting ZHC-001 compliance status;
- Detection of a system state deviation in any module operating under ZHC-001 oversight;
- Receipt of an on-demand ZHC-001 verification request via the Governance API Interface;
- Scheduled ZHC-001 compliance verification cycle initiation (per Section 8.2).

4.1.2 All ZHC-001 Verification Events are assigned a unique Event Reference Number (ERN) in the format `ZHC001-AVE-[YYYY]-[NNNNNN]` and are recorded in the ATLM ZHC-001 audit partition within 500 milliseconds of event trigger.

4.2 Real-Time ZHC-001 Compliance Verification Procedures

4.2.1 Upon trigger of a ZHC-001 Verification Event, the ZIL shall:

1. Authenticate the event source against the current ZHC-001 authorized entity registry;

2. Retrieve the current ZHC-001 policy snapshot and verify its integrity hash against the policy registry master;
3. Submit the event parameters to the CVE for deterministic compliance assessment;
4. Receive and validate the CVE compliance signal;
5. Generate a ZHC-001 compliance verdict (COMPLIANT, NON-COMPLIANT, PENDING, or INDETERMINATE);
6. Transmit the verdict to the ACG for certificate generation and to the ATLM for log entry.

4.2.2The target processing latency for a ZHC-001 Verification Event from trigger to verdict is 2,000 milliseconds or less under standard operational conditions. Latency breaches are logged as Severity 3 anomalies.

4.3 ZHC-001 Override and Escalation Handling

4.3.1A ZHC-001 override condition arises when a ZHC-001 Verification Event produces a NON-COMPLIANT or INDETERMINATE verdict, or when the ZIL detects a policy integrity failure. In such cases, Astra-Veritas shall:

7. Immediately raise an Anomaly Flag at a minimum Severity 2;
8. Suspend processing of dependent governance events pending resolution;
9. Notify the ZHC-001 Compliance Lead and the Chief Technology Governance Officer via the designated escalation channel;
10. Issue a Provisional Attestation Certificate with status SUSPENDED and document the override condition in the ATLM ZHC-001 partition.

4.3.2No ZHC-001 override condition may be resolved without explicit written authorization from the ZHC-001 Compliance Lead, countersigned by the Chief Technology Governance Officer.

4.4 ZHC-001 Verification Output Formats

4.4.1ZHC-001 verification outputs shall be structured as follows:

Output Field	Format	Description
Event Reference Number (ERN)	ZHC001-AVE-[YYYY]-[NNNNNN]	Unique identifier for the ZHC-001 Verification Event
Verification Timestamp	ISO 8601 UTC	Date and time of verdict generation
Compliance Verdict	COMPLIANT / NON-COMPLIANT / PENDING / INDETERMINATE	Primary compliance status determination
Policy Snapshot Hash	SHA-256 hex string	Integrity hash of the ZHC-001 policy version applied
Policy Version Reference	ZHC-001-v[N.N.N]	Version of ZHC-001 applied to the verification
Attestation Certificate ID	AV-CERT-[TYPE]-[YYYY]-[NNNNNN]	Reference to the generated Attestation Certificate
Anomaly Flag Reference	AV-ANO-[SEVERITY]-[NNNNNN] or NULL	Anomaly flag reference if raised; NULL if none

5. Verification Protocols

5.1 Document Integrity Verification

5.1.1 Document Integrity Verification (DIV) is the process by which Astra-Veritas confirms that a governance document has not been altered since its last authorized version was recorded. The CVE computes an Integrity Hash of the submitted document and compares it against the registered hash in the NSTT document registry. A match constitutes a COMPLIANT signal; any deviation constitutes a NON-COMPLIANT signal and triggers an Anomaly Flag.

5.1.2 All documents bearing cross-reference to ZHC-001, NSTT-GOV-SOV-001, NSTT-GOV-LEGAL-001, or NSTT-GOV-AUD-001 are subject to mandatory DIV upon submission to any NSTT governance process.

5.2 Process Compliance Verification

5.2.1 Process Compliance Verification (PCV) confirms that a completed or in-progress NSTT governance process has been executed in accordance with the applicable procedural governance standard. The CVE evaluates the process event

stream against the authorized process definition and identifies any procedural deviations.

5.2.2PCV outputs include a step-level compliance map identifying each process step as COMPLIANT, NON-COMPLIANT, or SKIPPED, together with an overall process compliance verdict.

5.3 Personnel Identity Attestation

5.3.1Personnel Identity Attestation (PIA) verifies the identity and access authorization of individuals performing governance actions within the NSTT environment. The CVE validates identity tokens against the NSTT authorized personnel registry and confirms that the individual's access classification is sufficient for the action performed.

5.3.2Identity attestation records are embedded in all Attestation Certificates relating to actions performed by identified personnel, creating a verifiable chain of personal accountability.

5.4 System State Verification

5.4.1System State Verification (SSV) confirms that NSTT technical systems are operating within their authorized configuration baselines. The CVE compares live system state snapshots against integrity-hashed baseline records and reports any deviation as a compliance signal.

5.4.2SSV is conducted automatically at the intervals specified in Section 8.2 and on demand per Section 8.3. Critical system state deviations are escalated per the Anomaly Classification Matrix (Section 7.1).

5.5 Cross-Module Verification (Blue Bird, Moonlight Integration)

5.5.1Astra-Veritas performs Cross-Module Verification (CMV) to attest the integrity of data and process exchanges between NSTT governance modules. CMV is triggered automatically for all designated inter-module exchanges involving Blue Bird and Moonlight.

5.5.2For Blue Bird integration, CMV verifies the integrity and authorization of strategic operation records submitted for governance attestation. For Moonlight integration, CMV confirms that data classification, sovereignty tagging, and cross-

boundary transfer authorizations are valid and ZHC-001 compliant at the point of exchange.

5.5.3All CMV events produce a Cross-Module Attestation Certificate referencing both the originating module and the Astra-Veritas verification record.

6. Attestation Certificate Framework

6.0.1The Attestation Certificate Framework defines the classification, issuance conditions, revocation procedures, and chain of custody requirements for all certificates produced by the ACG.

6.1 Certificate Types and Classifications

Certificate Type	Code	Scope	ZHC-001 Tag	Validity Period
Governance Compliance Certificate	GCC	Full process or policy compliance attestation	Required if event is ZHC-001 tagged	Until superseded
Document Integrity Certificate	DIC	Point-in-time document integrity attestation	Required for ZHC-001 documents	Perpetual (tied to document version)
Personnel Identity Certificate	PIC	Individual identity and authorization attestation	Required for ZHC-001 events	Session-bound
System State Certificate	SSC	System configuration and state attestation	Mandatory for ZHC-001 systems	Until next SSV cycle
Cross-Module Attestation Certificate	CMAC	Inter-module data/process exchange attestation	Mandatory for all CMV events	Until superseded by next exchange
ZHC-001 Attestation Certificate	ZAC	Dedicated ZHC-001 compliance event certificate	Mandatory - ZHC-001 reference embedded	As defined in ZHC-001
Provisional Attestation	PAC	Interim attestation	Issued in override	72 hours or until resolved

Certificate Type	Code	Scope	ZHC-001 Tag	Validity Period
Certificate		during override or escalation	conditions	
Audit Evidence Certificate	AEC	Packaged evidence submission for AUD-001 proceedings	As applicable	Perpetual (archive-grade)

6.2 Certificate Issuance Conditions

6.2.1A certificate shall be issued only when all of the following conditions are satisfied: (a) the CVE has produced a definitive compliance signal; (b) the ZIL has returned a ZHC-001 verdict where required; (c) the ACG has verified that the requesting entity is authorized to receive the certificate type requested; and (d) the ATLM has confirmed receipt of the pre-issuance log entry.

6.3 Certificate Revocation Procedures

6.3.1A certificate may be revoked upon: discovery of fraudulent input data; identification of a rule set defect affecting the certificate's validity; or written directive from the Chief Technology Governance Officer. Revocation is recorded in the certificate revocation registry, and all downstream consumers of the certificate are notified via the GAI within 30 minutes of revocation.

6.4 Certificate Chain of Custody

6.4.1Every Attestation Certificate maintains a chain of custody record embedded in its metadata, including: the originating verification event ERN; the CVE rule set version applied; the ZHC-001 policy version (where applicable); the issuing ACG instance identifier; and the ATLM log entry reference. This chain of custody is cryptographically sealed at the time of issuance and is immutable thereafter.

7. Anomaly Detection & Response

7.1 Anomaly Classification Matrix

Severity	Anomaly Type	Examples	Automated Response	Escalation Path	Resolution SLA
Severity 1 - Critical	Integrity Breach / ZIL Session Failure	Integrity hash mismatch on ZHC-001 document; ZIL policy registry authentication failure	Immediate process suspension; Provisional Certificate issued; ZHC-001 verification halt	ZHC-001 Compliance Lead → CTGO → Board Governance Committee	4 hours
Severity 2 - High	ZHC-001 Non-Compliance / Override Condition	NON-COMPLIANT ZHC-001 verdict; INDETERMINATE verdict on mandatory event; unauthorized access attempt	Dependent event suspension; escalation notification dispatched; investigation dossier initiated	Astra-Veritas Module Lead → ZHC-001 Compliance Lead → CTGO	24 hours
Severity 3 - Medium	Performance Degradation / Process Deviation	Latency breach on ZHC-001 event; process step NON-COMPLIANT; system state minor deviation	Alert dispatched; affected verification cycle flagged; increased monitoring activated	Astra-Veritas Module Lead → Chief Audit Executive	72 hours
Severity 4 - Low	Configuration Drift / Minor Deviation	Non-critical system parameter outside baseline; certificate delivery latency; log backlog	Log entry created; scheduled review queue added	Astra-Veritas Operations Team	Next scheduled maintenance cycle
Severity 5 - Informational	Advisory / Monitoring Notification	Approaching performance threshold; scheduled	Informational log entry; dashboard update	Astra-Veritas Operations Team	No mandatory SLA

Severity	Anomaly Type	Examples	Automated Response	Escalation Path	Resolution SLA
		review reminder; policy expiry notice			

7.2 Real-Time Alert Protocol

7.2.1 Upon detection of a Severity 1 or Severity 2 anomaly, the ADS shall dispatch an immediate alert to all designated recipients via the NSTT secure notification channel within 60 seconds of detection. The alert shall include: the Anomaly Flag Reference, severity classification, affected Event Reference Number(s), automated response actions taken, and the assigned investigation authority.

7.2.2 Severity 3 alerts shall be dispatched within 15 minutes of detection. Severity 4 and 5 notifications shall be included in the next scheduled operational report cycle.

7.3 Anomaly Investigation Procedures

7.3.1 Upon receipt of an anomaly investigation assignment, the designated investigation authority shall: (a) review the ATLM log entries for the affected event window; (b) retrieve all Attestation Certificates associated with the anomaly period; (c) assess the root cause of the anomaly against the CVE rule set and ZHC-001 parameters; and (d) produce a written Investigation Report within the applicable resolution SLA.

7.4 Remediation and Re-Attestation

7.4.1 Following resolution of an anomaly, the investigation authority shall authorize a re-verification run for all affected events. Astra-Veritas shall reprocess the affected verification events and issue replacement Attestation Certificates referencing the original anomaly flag and the remediation record. Superseded certificates are automatically revoked per Section 6.3.

8. Operational Procedures

8.1 Module Initialization and Readiness Check

8.1.1 Prior to entering operational status, Astra-Veritas shall execute a full initialization sequence including: (a) CVE rule set integrity verification; (b) ZIL policy registry authentication and snapshot retrieval; (c) ATLM log integrity chain verification; (d) ADS baseline calibration; (e) GAI endpoint health check; and (f) ACG certificate signing key validation. Module operational status is confirmed only upon successful completion of all six checks.

8.1.2 Any failed initialization check is classified as a Severity 2 anomaly and shall prevent the module from entering operational status until resolved.

8.2 Daily Verification Run Schedule

8.2.1 Astra-Veritas shall execute scheduled verification runs in accordance with the following cycle:

Run Type	Frequency	UTC Start Time	Scope
Full Governance Compliance Scan	Daily	02:00 UTC	All active governance artifacts and process records
ZHC-001 Compliance Cycle	Daily	04:00 UTC	All ZHC-001 tagged events and policy parameters
System State Verification	Every 6 hours	00:00 / 06:00 / 12:00 / 18:00 UTC	All NSTT module system state baselines
Cross-Module Exchange Audit	Daily	06:00 UTC	All Blue Bird and Moonlight exchange records (prior 24 hours)
ATLM Log Integrity Verification	Daily	03:00 UTC	Full audit log chain integrity check

8.3 On-Demand Verification Requests

8.3.1 Authorized requestors may submit on-demand verification requests through the GAI at any time during module operational hours. On-demand requests shall be processed within the target latency specified in Section 4.2.2 for ZHC-001 events,

and within 10,000 milliseconds for all other verification types under standard load conditions.

8.3.2On-demand requests are subject to the same access control restrictions specified in Section 11. All on-demand verification events are logged in the ATLM regardless of outcome.

8.4 Emergency Verification Protocol

8.4.1The Emergency Verification Protocol (EVP) is activated by written directive from the CTGO or ZHC-001 Compliance Lead in response to a Severity 1 anomaly, a sovereign governance crisis, or a declared NSTT emergency. Under the EVP, Astra-Veritas shall: suspend all non-essential verification runs; prioritize processing of emergency-designated events; activate enhanced ATLM logging at maximum resolution; and dispatch continuous status updates to the CTGO at 30-minute intervals until the EVP is formally stood down.

8.5 Module Shutdown and Continuity

8.5.1Planned shutdown of Astra-Veritas shall require 48-hour advance notification to the CTGO and Chief Audit Executive, a pre-shutdown ATLM log integrity seal, and a documented continuity arrangement confirming that no ZHC-001 verification events are scheduled during the shutdown window. Emergency shutdown may be authorized by the CTGO without advance notice, with a post-shutdown log entry required within 2 hours of restoration.

9. Governance Integration Points

9.0.1Astra-Veritas maintains formal integration interfaces with the following NSTT governance frameworks and modules:

Integration Target	Document Reference	Integration Type	Astra-Veritas Role	Output to Target
Sovereignty Verification Framework	NSTT-GOV-SOV-001	Bi-directional authenticated API	Primary verifier of sovereignty boundary compliance events	ZAC, GCC, Compliance Signals

Integration Target	Document Reference	Integration Type	Astra-Veritas Role	Output to Target
Legal Compliance Attestation	NSTT-GOV-LEGAL-001	Attestation delivery (push)	Issuer of legally admissible attestation certificates	AEC, GCC, Investigation Reports
Audit Evidence Generation	NSTT-GOV-AUD-001	Bi-directional authenticated API; log export	Primary source of audit evidence and verification records	AEC, ATLM exports, Anomaly Reports
Blue Bird (Strategic Operations)	NSTT-GOV-MASTER-001 (Blue Bird module)	Cross-module verification (CMV)	Verifier of strategic operation record integrity and authorization	CMAC, SSC, Anomaly Flags
Moonlight (Data Sovereignty)	NSTT-GOV-MASTER-001 (Moonlight module)	Cross-module verification (CMV)	Verifier of data classification, sovereignty tagging, and transfer authorization	CMAC, ZAC, DIC
Master Governance Framework	NSTT-GOV-MASTER-001	Policy snapshot subscription; reporting	Consumer of master policy parameters; reporter of module status	Module status reports, GCC, Compliance dashboards

10. Performance & Reliability Standards

10.1 Verification Throughput Requirements

10.1.1 Astra-Veritas shall maintain a minimum verified throughput of 500 verification events per hour under standard operational load, and a minimum of 200 verification events per hour under degraded conditions (defined as the activation of fallback subsystems). ZHC-001 Verification Events shall be afforded processing priority over all other event types during periods of high load.

10.2 Availability and Uptime Obligations

Service Tier	Function	Uptime Requirement	Maximum Allowable Downtime (per year)	Measurement Window
Tier 1 – ZHC-001 Processing	ZIL, CVE (ZHC-001 events)	99.95%	4.38 hours	Calendar year
Tier 2 – Core Verification	CVE (all other events), ACG, GAI	99.9%	8.76 hours	Calendar year
Tier 3 – Anomaly Detection	ADS real-time monitoring	99.9%	8.76 hours	Calendar year
Tier 4 – Audit Logging	ATLM append operations	99.99%	0.88 hours	Calendar year
Tier 5 – Reporting & Dashboards	Compliance dashboards, report generation	99.5%	43.8 hours	Calendar year

10.3 Error Tolerance and Fallback Procedures

10.3.1 In the event of a primary subsystem failure, Astra-Veritas shall automatically engage the designated fallback instance within 120 seconds. All verification events in-flight at the time of failure are resubmitted to the fallback instance. Events that cannot be recovered are logged as INDETERMINATE and flagged for manual review under the Anomaly Classification Matrix at Severity 2.

10.4 ZHC-001 Performance Compliance Thresholds

10.4.1 ZHC-001 performance compliance thresholds are as follows: (a) end-to-end ZHC-001 event processing latency must not exceed 2,000 milliseconds for 99% of events in any 24-hour window; (b) ZHC-001 certificate issuance must complete within 500 milliseconds of verdict generation; and (c) ZHC-001 ATLM log entry must be recorded within 500 milliseconds of event trigger. Sustained breach of any threshold over a 48-hour window constitutes a reportable performance failure under NSTT-GOV-AUD-001.

11. Access Control & Security

11.1 Module Access Hierarchy

Access Level	Role	Permitted Functions
Level 1 – Full Administrative	Chief Technology Governance Officer	All functions including EVP activation, module shutdown authorization, override resolution
Level 2 – Module Governance	Astra-Veritas Module Lead; ZHC-001 Compliance Lead	Verification run management, certificate management, anomaly investigation, ZHC-001 escalation
Level 3 – Audit Authority	Chief Audit Executive; NSTT-GOV-AUD-001 designated auditors	Read-only ATLM access; AEC retrieval; audit report request submission
Level 4 – Operations	Astra-Veritas Operations Team	Scheduled run monitoring, Severity 4-5 anomaly response, on-demand request facilitation
Level 5 – Consumer Read	Authorized NSTT module systems (Blue Bird, Moonlight, SOV-001)	GAI-mediated certificate retrieval; compliance signal query; status endpoint polling

11.2 Authentication Requirements

11.2.1 All access to Astra-Veritas functions requires multi-factor authentication (MFA) via NSTT-approved authentication credentials. Module-to-module API access requires mutual TLS certificate authentication with NSTT-issued certificates. All authentication events are logged in the ATLM.

11.3 Data Handling and Classification

11.3.1 All data processed by Astra-Veritas is classified at a minimum of RESTRICTED. Data containing ZHC-001 event information is classified RESTRICTED – ZHC-001 CONTROLLED and is subject to the handling requirements of ZHC-001 and NSTT-GOV-MASTER-001. No Astra-Veritas data may be transmitted outside NSTT-controlled infrastructure without explicit CTGO authorization.

11.4 Security Incident Response

11.4.1 Any suspected security incident affecting Astra-Veritas shall be reported to the CTGO and ZHC-001 Compliance Lead immediately upon detection. The ADS shall automatically classify unauthorized access attempts as Severity 2 anomalies. Security incidents affecting ZHC-001 processing are subject to the ZHC-001 override and escalation procedures of Section 4.3 in addition to NSTT security incident response procedures.

12. Reporting & Outputs

12.1 Standard Output Formats and Schedules

12.1.1 Astra-Veritas shall produce the following standard reports on the schedule indicated:

- **Daily Verification Summary Report:** Produced at 08:00 UTC each day; covers all verification events from the preceding 24-hour window; distributed to Level 1-3 access roles.
- **ZHC-001 Compliance Status Report:** Produced daily at 08:30 UTC; covers all ZHC-001 Verification Events; distributed to the CTGO and ZHC-001 Compliance Lead.
- **Weekly Attestation Register:** Produced every Monday at 09:00 UTC; lists all Attestation Certificates issued, revoked, or suspended in the preceding week.
- **Monthly Governance Performance Report:** Produced on the first business day of each month; includes throughput statistics, uptime measurement, anomaly summaries, and ZHC-001 performance compliance assessment.
- **Bi-Annual Governance Review Package:** Produced at each review cycle; supports formal module review per the review cycle specified in the Document Control table.

12.2 Exception Reports

12.2.1 Exception reports are generated automatically upon the occurrence of any Severity 1 or 2 anomaly, any ZHC-001 override condition, any certificate revocation, or any SLA breach. Exception reports are dispatched within 30 minutes of the triggering event to the appropriate escalation recipients.

12.3 ZHC-001 Compliance Dashboards

12.3.1 Astra-Veritas maintains a real-time ZHC-001 compliance dashboard accessible to Level 1-2 access role holders. The dashboard displays current ZHC-001 compliance status, active verification events, recent ZAC issuances, open anomaly flags, and ZIL session health indicators.

12.4 Integration with NSTT-GOV-AUD-001 Reporting

12.4.1 All Astra-Veritas standard and exception reports, together with ATLM log exports and Audit Evidence Certificates, are made available to NSTT-GOV-AUD-001 through the designated audit evidence interface. AUD-001 audit proceedings may invoke on-demand evidence packages through the GAI at Level 3 access authorization.

13. Module Maintenance & Versioning

13.1 Update and Patching Procedures

13.1.1 All updates to the Astra-Veritas module, including CVE rule set updates, ZIL policy parameter updates, ACG certificate classification changes, and ADS baseline recalibrations, shall be processed through the formal NSTT change management procedure. No update shall be applied to a production instance without prior written authorization from the CTGO and confirmation of successful testing per Section 13.3.

13.1.2 Emergency security patches may be applied with CTGO verbal authorization, subject to written confirmation within 4 hours and post-application validation within 24 hours.

13.2 Version Control Requirements

13.2.1 Astra-Veritas version numbers follow the format [MAJOR].[MINOR].[PATCH]. A MAJOR increment is required for any change to the CVE verification logic or ZIL integration protocol. A MINOR increment is required for changes to certificate classification rules, anomaly detection thresholds, or operational procedures. A PATCH increment is required for security fixes and non-functional corrections. All version changes require a corresponding update to this specification document.

13.3 Testing and Validation Standards

13.3.1 Prior to deployment of any module update, the following validation suite must be completed in a non-production environment: (a) full determinism verification confirming identical outputs for identical inputs; (b) ZHC-001 integration test battery covering all defined trigger conditions; (c) anomaly detection test suite covering all five severity classifications; (d) performance load test confirming throughput and latency requirements under peak load; and (e) ATLM integrity chain continuity test. All test results must be retained as Audit Evidence Certificates.

13.4 Rollback Procedures

13.4.1 In the event that a deployed update causes a Severity 1 or 2 anomaly, the Module Lead shall initiate an immediate rollback to the last verified stable version. The rollback shall be completed within 2 hours of authorization. Following rollback, a full initialization sequence (Section 8.1) shall be executed, and a post-incident report shall be produced within 24 hours detailing the root cause and remediation plan.

14. Attestation & Authorization

AUTHORIZATION STATEMENT

The undersigned hereby attest that this specification document (NSTT-MOD-AV-001, Version 1.0.0) has been reviewed, validated, and approved for publication and operational effect as of the effective date stated herein. This document

constitutes the authoritative governance specification for the Astra-Veritas module within the NSTT governance suite and shall be treated in accordance with its RESTRICTED classification under NSTT-GOV-MASTER-001.

Chief Technology Governance Officer

NSTT Office of Technology Governance

Date: _____



Astra-Veritas Module Lead

NSTT Office of Technology Governance – Astra-Veritas Programme

Date: _____



ZHC-001 Compliance Lead

NSTT ZHC-001 Compliance Authority

Date: _____



Chief Audit Executive

NSTT Office of Internal Audit – NSTT-GOV-AUD-001

Date: _____



NSTT-MOD-AV-001 | Astra-Veritas Module Specification | Version 1.0.0 |

Effective: 20 May 2026

RESTRICTED – MODULE SPECIFICATION CONTROLLED | NSTT Office of Technology
Governance

ZHC Integration Reference: ZHC-001 | Cross-References: NSTT-GOV-SOV-001,
NSTT-GOV-LEGAL-001, NSTT-GOV-AUD-001, NSTT-GOV-MASTER-001

RESTRICTED – MODULE SPECIFICATION CONTROLLED